

EDUSAMBAM ASSESSMENT SERIES

Practice Test: Sets & Matrices

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Subject	Mathematics (Maths)	Topic	Practice Test: Sets & Matrices	Total Marks	100
Total Questions	45	Time Allowed	1 Hour (60 minutes)	Version	A

GENERAL INSTRUCTIONS

1. Read each question carefully before answering.
2. For Section A and Section B, mark one answer only.
3. For Sections C and E, write your answer clearly in the space provided.
4. For Section D, write the correct letter from Column B against each item in Column A.
5. Do not use correction fluid. Keep all answers clear and legible.
6. This worksheet is prepared from the EDUSAMBAM practice assessment supplied for this paper.

PAPER PATTERN

Section	Questions	Marks Each	Total
Section A — Multiple Choice	15	2	30
Section B — True / False	10	2	20
Section C — Fill in the Blank	10	2	20
Section D — Column Matching	5 pairs	2	10
Section E — Short Answer	5	4	20
Total	45	—	100

Section A — Multiple Choice

30 marks

1. The symbol $\in$ means:

Basic

- A. Is a subset of
- C. Union

- B. Is not equal to
- D. Is an element of

Answer: ☐ A ☐ B ☐ C ☐ D

2. $n(A)$ represents:

Basic

- A. The complement of A
- C. The universal set

- B. The number of elements in A
- D. The empty set

Answer: ☐ A ☐ B ☐ C ☐ D

3. The universal set is written:

Basic

- A. ξ
- C. $\in$

- B. $\emptyset$
- D. $\subset$

Answer: ☐ A ☐ B ☐ C ☐ D

4. $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4, 5\}$. Is $A \subset B$?

Intermediate

- A. No
- C. Yes

- B. Cannot be determined
- D. Only if $A = B$

Answer: ☐ A ☐ B ☐ C ☐ D

5. $A = \{2, 4, 6\}$, $B = \{4, 6, 8\}$. Find $A \cup B$.

Basic

- A. $\{2, 4, 6, 8\}$
- C. $\{2, 8\}$

- B. $\{4, 6\}$
- D. $\{2, 4, 4, 6, 6, 8\}$

Answer: ☐ A ☐ B ☐ C ☐ D

6. $A = \{2, 4, 6\}$, $B = \{4, 6, 8\}$. Find $A \cap B$.

Basic

- A. $\{2, 4, 6, 8\}$
- C. $\{4, 6\}$

- B. $\{2\}$
- D. $\{8\}$

Answer: ☐ A ☐ B ☐ C ☐ D

7. The counting formula for two sets is:

Intermediate

- A. $n(A \cup B) = n(A) + n(B)$
- C. $n(A \cup B) = n(A) \times n(B)$

- B. $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- D. $n(A \cup B) = n(A \cap B)$

Answer: ☐ A ☐ B ☐ C ☐ D

8. The order of a matrix with 2 rows and 3 columns is written:

Basic

- A. 3×2
- C. 5

- B. 6
- D. 2×3

Answer: ☐ A ☐ B ☐ C ☐ D

9. Two matrices can only be added if:

Basic

- A. They are both square
- C. They have the same order

- B. They both have determinant 0
- D. Neither has any negative numbers

Answer: ☐ A ☐ B ☐ C ☐ D

10. Matrix multiplication is performed using:

Intermediate

- A. Row times column, then sum the products
- C. Adding corresponding elements

- B. Element-by-element multiplication
- D. Dividing corresponding elements

Answer: ☐ A ☐ B ☐ C ☐ D

11. $A = \begin{pmatrix} 1 & 0 \\ 2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$. Find the top-left element of AB .

Advanced

- A. 0
- C. 3

- B. 1
- D. 2

Answer: ☐ A ☐ B ☐ C ☐ D

12. Matrix multiplication is:

Intermediate

- A. Always commutative
C. The same as addition

- B. Generally NOT commutative
D. Only defined for square matrices

Answer: ☐ A ☐ B ☐ C ☐ D

13. The determinant of $\begin{pmatrix} 3 & 2 \\ 1 & 4 \end{pmatrix}$ is:

Intermediate

- A. 14
C. 2

- B. 10
D. 12

Answer: ☐ A ☐ B ☐ C ☐ D

14. A matrix with a determinant of zero is called:

Intermediate

- A. An identity matrix
C. Invertible

- B. Symmetric
D. Singular

Answer: ☐ A ☐ B ☐ C ☐ D

15. To find the inverse of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the formula is:

Advanced

- A. $\begin{pmatrix} d & b \\ c & a \end{pmatrix}$
C. $\frac{1}{\det(A)} \times \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

- B. $\det(A) \times \begin{pmatrix} a & b \\ c & d \end{pmatrix}$
D. $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ itself

Answer: ☐ A ☐ B ☐ C ☐ D

Section B — True / False

20 marks

16. The empty set is written $\emptyset$ or $\{\}$.

Basic

☐ True ☐ False

17. Every set is a subset of the empty set.

Intermediate

☐ True ☐ False

18. $A \cup B$ contains elements in A, B, or both.

Basic

☐ True ☐ False

19. $A \cap B$ contains elements only in A, not B.

Basic

☐ True ☐ False

20. $n(A')$ means the number of elements NOT in A, within ξ .

Intermediate

☐ True ☐ False

21. Two matrices of different orders can always be multiplied together.

Intermediate

☐ True ☐ False

22. The identity matrix behaves like the number 1 in ordinary multiplication.

Basic

☐ True ☐ False

23. A matrix with a non-zero determinant has an inverse.

Intermediate

☐ True ☐ False

24. Matrix addition requires multiplying corresponding elements.

Basic

☐ True ☐ False

25. Simultaneous equations written as $AX = B$ can be solved using $X = A^{-1}B$.

Intermediate

☐ True ☐ False

Section C — Fill in the Blank

20 marks

26. The symbol for "is an element of" is _____.

Answer:

27. A set with no elements is called the _____ set.

Answer:

28. Elements common to both A and B form the _____ of A and B.

Answer:

29. The complement of A, written A' , contains elements not in A but in the _____ set.

Answer:

30. Filling a Venn diagram from a word problem should start with the most _____ region.

Answer:

31. The order of a matrix is given as rows $\times$ _____.

Answer:

32. Multiplying every element of a matrix by a number is called _____ multiplication.

Answer:

33. The determinant of a 2×2 matrix (a b over c d) is $ad -$ _____.

Answer:

34. A matrix with a determinant of zero has no _____.

Answer:

35. The matrix that leaves any compatible matrix unchanged when multiplied is called the _____ matrix.

Answer:

Section D — Column Matching

10 marks

36–40. Match each term to its correct definition.

Column A	Ans.	Column B
Union	_____	A. Elements in ξ that are not in A
Intersection	_____	B. Elements in A, B, or both
Complement	_____	C. Leaves any compatible matrix unchanged when multiplied
Determinant	_____	D. Elements in both A and B
Identity matrix	_____	E. $ad - bc$ for a 2×2 matrix

Section E — Short Answer

20 marks

41. Explain how to use the counting formula $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ in a survey problem.

42. Describe the order in which you should fill in a three-set Venn diagram from a word problem.

43. Explain the difference between matrix addition and matrix multiplication.

44. Describe how to find the determinant and inverse of a 2×2 matrix.

45. Explain how matrices can be used to solve a pair of simultaneous equations.

Candidate Signature: _____

Invigilator Signature: _____

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Marks Obtained: _____ / 100 Checked By: _____

Result / Grade: _____ Date Checked: _____

Practice Test: Sets & Matrices — Answer Key

Teacher reference: equivalent grammatically correct wording is acceptable for open-ended items when the meaning is preserved.

Section A — Multiple Choice

- | | |
|------|-------|
| 1. D | |
| 2. B | 9. C |
| 3. A | 10. A |
| 4. C | 11. D |
| 5. A | 12. B |
| 6. C | 13. B |
| 7. B | 14. D |
| 8. D | 15. C |

Section B — True / False

- | | |
|-----------|-----------|
| 16. True | 21. False |
| 17. False | 22. True |
| 18. True | 23. True |
| 19. False | 24. False |
| 20. True | 25. True |

Section C — Fill in the Blank

- | | |
|------------------|--------------|
| 26. $\in$ | 31. columns |
| 27. empty | 32. scalar |
| 28. intersection | 33. bc |
| 29. universal | 34. inverse |
| 30. overlapping | 35. identity |

Section D — Column Matching

- | | |
|----------------------|-------------------------|
| 36. Union — B | |
| 37. Intersection — D | 39. Determinant — E |
| 38. Complement — A | 40. Identity matrix — C |

Section E — Short Answer (Model Answer Keywords)

41. subtract; intersection; double counted
42. most overlapping; specific region; outward
43. corresponding elements; row times column; different methods
44. $ad-bc$; swap; negate; divide
45. $AX=B$; $AX = B$; inverse; inverse matrix